Example. Let's first preform the calculation for a normal particle. Generalising the result of $S_z|(0,k)\rangle$ from the above (notice how we never used the fact that the momentum of the arbitrary state was 0 up to the second last line), we have

$$S_j|(p,i)\rangle = \frac{1}{2E_p}\sum_{k=1}^{2} b^k(p)^\dagger u_+^k(p)^\dagger \left(\frac{1}{2}\Sigma^j\right) u_+^i(p)|0\rangle$$

The helicity operator can then be written as

$$\begin{aligned}\sum_{j=1}^{3} n_j S_j|(p,i)\rangle =& \sum_{j=1}^{3} n_j \frac{1}{2E_p}\sum_{k=1}^{2} b^k(p)^\dagger u_+^k(p)^\dagger \left(\frac{1}{2}\Sigma^j\right) u_+^i(p)|0\rangle \\ =& \frac{1}{2E_p}\sum_{k=1}^{2} b^k(p)^\dagger u_+^k(p)^\dagger \Big[\sum_{j=1}^{3}\frac{1}{2}n_j\Sigma^j u_+^i(p)\Big]|0\rangle\end{aligned}$$

where n_j is again defined as $p^j/|p|$. We can now employ the results from the previously problem, where

$$\begin{aligned}\sum_j \frac{1}{2}n_j\Sigma^j u_+^i(p) =& \sum_j \frac{1}{2}\begin{pmatrix} n_j\sigma^j & 0 \\ 0 & n_j\sigma^j\end{pmatrix}\begin{pmatrix}\xi^{\pm}(p) \\ \xi^{\pm}(p)\end{pmatrix} = \sum_j \frac{1}{2}\begin{pmatrix} n_j\sigma^j\xi^{\pm}(p) \\ n_j\sigma^i\xi^{\pm}(p)\end{pmatrix} \\ =& \pm\frac{1}{2}\begin{pmatrix}\xi^{\pm}(p) \\ \xi^{\pm}(p)\end{pmatrix} = \pm\frac{1}{2}u_+^i(p)\end{aligned}$$

(the $+$ sign for $i=1$ and $-$ for $i=2$). Substituting this into the above, we obtain

$$H|(p,i)\rangle = \frac{1}{2E_p}\sum_{k=1}^{2} b^k(p)^\dagger u_+^k(p)^\dagger\Big[\pm\frac{1}{2}u_+^i(p)\Big] = \sum_{k=1}^{2} b^k(p)^\dagger\Big[\pm\frac{1}{2}\Big]\delta^{ki}|0\rangle = \pm\frac{1}{2}b^i(p)^\dagger|0\rangle = \pm\frac{1}{2}|(p,i)\rangle$$

which is exactly the result we expected! Similarly,

$$H|(p,i')\rangle = \frac{1}{2E_p}\sum_{k=1}^{2} d^k(p)^\dagger u_-^k(p)^\dagger\Big[\pm\frac{1}{2}u_-^{i'}(p)\Big] = \sum_{k=1}^{2} -d^k(p)^\dagger\Big[\pm\frac{1}{2}\Big]\delta^{ki'}|0\rangle = \mp\frac{1}{2}d^{i'}(p)^\dagger|0\rangle = \mp\frac{1}{2}|(p,i')$$